

7.2 Transverse Waves: EM Spectrum & Polarisation

Question Paper

Course	CIE A Level Physics
Section	7. Waves
Topic	7.2 Transverse Waves: EM Spectrum & Polarisation
Difficulty	Easy

Time allowed: 20

Score: /10

Percentage: /100

Question 1

The following are statements about waves. Which statement below describes a situation in which polarisation would not happen?

- A. microwaves pass through a metal grid
- B. sound waves pass through a metal grid
- C. light waves are reflected
- D. light waves are scattered

[1 mark]

Question 2

A student looked at the liquid crystal display on their calculator with a polarising film. They observed that as they rotated the film, the display changed.

Which property of the radiation from the calculator display is described correctly?

- A. the emitted radiation is a transverse wave
- B. the emitted radiation is a wave with 3 cm wavelength
- C. the emitted radiation is unpolarised
- D. the emitted radiation is a longitudinal wave

[1 mark]

Question 3

Which of the following is the approximate range of frequencies of infrared radiation?

- A. $1 \times 10^{14} - 1 \times 10^{16}$ Hz
- B. $1 \times 10^{12} - 1 \times 10^{14}$ Hz
- C. $1 \times 10^9 - 1 \times 10^{11}$ Hz
- D. $1 \times 10^3 - 1 \times 10^9$ Hz

[1 mark]

Question 4

Which of the following waves can be polarised?

- A. transverse
- B. progressive
- C. stationary
- D. longitudinal

[1 mark]

Question 5

Which of the following statement is a correct description of part of the electromagnetic spectrum?

- A. waves of wavelength 5×10^{-7} m are infrared waves
- B. waves of wavelength 5×10^{-9} m are high energy gamma rays
- C. waves of wavelength 9×10^{-7} m are light waves
- D. waves of wavelength 3×10^{-8} m are ultraviolet waves

[1 mark]

Question 6

Which of the following give the order of magnitude of frequencies of the visible part of the electromagnetic spectrum?

- A. 10^{15} Hz
- B. 10^{14} Hz
- C. 10^{13} Hz
- D. 10^{16} Hz

[1 mark]

Question 7

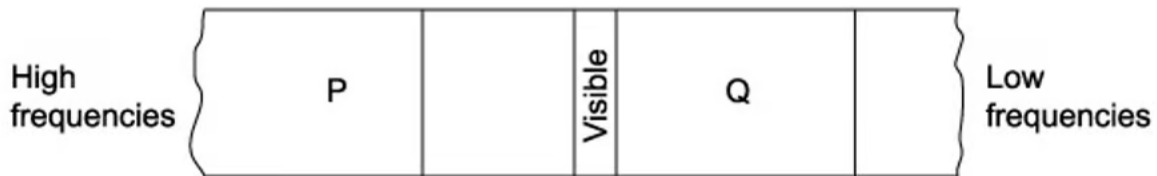
Malus's law shows the intensity of polarised light as it emerges through an analyser. Which is the correct equation to calculate the intensity?

- A. $I_0 \sin^2 \theta$
- B. $I_0 \cos^2 \theta$
- C. $I_0 \cos \theta$
- D. $\frac{I_0}{2} \cos^2 \theta$

[1 mark]

Question 8

The diagram shows part of the electromagnetic spectrum.



Which row correctly labels the regions marked P and Q?

	P	Q
A	infrared	x-ray
B	ultraviolet	x-ray
C	x-ray	infrared
D	infrared	microwaves

[1 mark]

Question 9

Which of the following do **not** travel at the speed of light in a vacuum?

- A. x-rays
- B. radio waves
- C. microwaves
- D. electrons

[1 mark]

Question 10

Which of the following is the number of wavelengths in one metre of visible light?

- A. 10^4
- B. 10^6
- C. 10^8
- D. 10^{10}

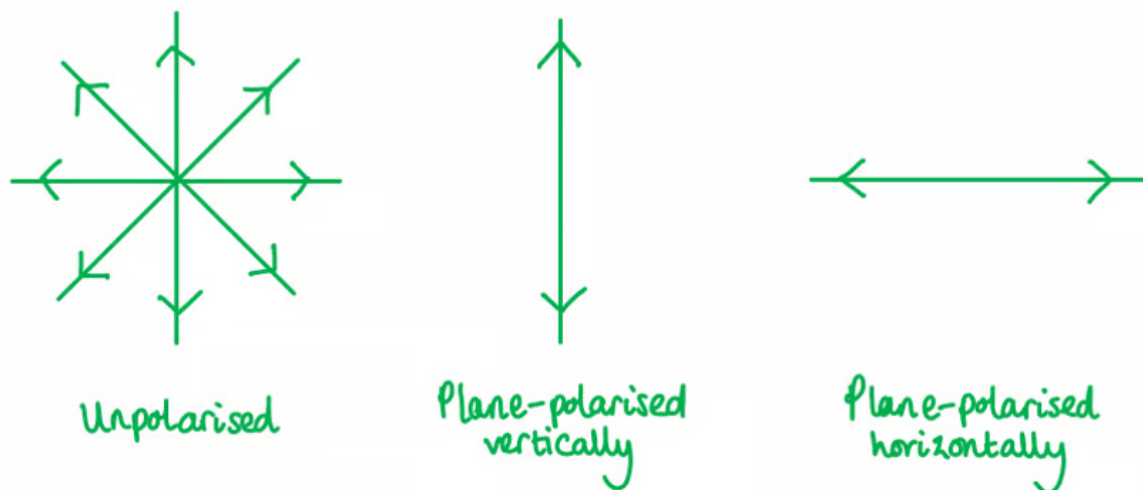
[1 mark]

Model Answers: Easy

1

The correct answer is **B** because:

- Polarisation is the process by which oscillations of transverse waves are made to occur in one plane only
- Transverse waves can be polarised as they can oscillate in many planes
- Sound, however, is a longitudinal wave
- Longitudinal waves cannot be polarised as they oscillate parallel to the direction of travel
- Therefore, sound waves would not be polarised by a metal grid



A is incorrect because microwaves are **transverse** waves so can be polarised, this can be done by passing the waves through a grid so that only the waves that fit through the slits can continue

C is incorrect because if light strikes a surface so that there is a 90° angle between the **reflected** and refracted rays, the **reflected** light will be linearly **polarised**

D is incorrect because the scattering of light off air molecules produces linearly polarised light in the plane perpendicular to the incident light

2

The correct answer is **A** because:

- Polarisation is the process by which oscillations of transverse waves are made to occur in one plane only
- Transverse waves can be polarised as they can oscillate in many planes
- The radiation emitted from the calculator's display is visible light
- Visible light is part of the electromagnetic spectrum, which is an example of a transverse wave

B is incorrect because a wave with a 3 cm wavelength would be a **microwave**

C is incorrect because as the display is changed as the film rotated this is evidence that the light emitted from the calculator is polarised

D is incorrect because a longitudinal wave can't be polarised as it oscillates parallel to the direction of travel

3

The correct answer is **B** because the range of **infrared** radiation is $3.0 \times 10^{11} - 4.3 \times 10^{14}$ Hz

A is incorrect because the **ultraviolet** light range is $7.5 \times 10^{14} - 3.0 \times 10^{16}$ Hz

C is incorrect because the **microwave** range is $3.0 \times 10^9 - 3.0 \times 10^{11}$ Hz

D is incorrect because the **radio** range is less than 3.0×10^9 Hz

4

The correct answer is **A** because:

- Polarisation is the process by which oscillations of transverse waves are made to occur in one plane only
- Transverse waves can be polarised as they can oscillate in many planes
- Light is an example of a transverse wave, and polarisation is a property which is unique to transverse waves only

B is incorrect because the definition of a **progressive** wave is one that moves **through** a material and **transfers energy**

C is incorrect because the definition of a **stationary wave** or **standing wave** is a combination of two waves moving in opposite directions each having the same amplitude and frequency so are in phase. The energy of the two waves cancel each other out. As the wave is not moving through the medium it can't be polarised

D is incorrect because a longitudinal wave can't be polarised as it **oscillates parallel** to the direction of travel

5

The correct answer is **D** because waves with a wavelength in the range of 4×10^{-7} to 10^{-8} m are **ultraviolet**

A is incorrect because waves with a wavelength in the range $4 - 7 \times 10^{-7}$ m are visible light

B is incorrect because waves with a wavelength in the range $10^{-8} - 4 \times 10^{-13}$ m are x-rays

C is incorrect because waves with a wavelength in the range $10^{-3} - 7 \times 10^{-7}$ are infrared

6

The correct answer is **B** because 10^{14} Hz is in the visible light range of $4.3 - 7.5 \times 10^{14}$ Hz

A is incorrect because 10^{15} Hz is in the **ultraviolet** light range: $7.5 \times 10^{14} - 3.0 \times 10^{16}$ Hz

C is incorrect because 10^{13} Hz is in the **infrared** range: $3.0 \times 10^{11} - 4.3 \times 10^{14}$ Hz

D is incorrect because 10^{16} Hz is in the **x-ray** range: $3.0 \times 10^{16} - 7.5 \times 10^{20}$ Hz

7

The correct answer is **B** because:

- Polarisation is the process by which the oscillations are made to occur in one

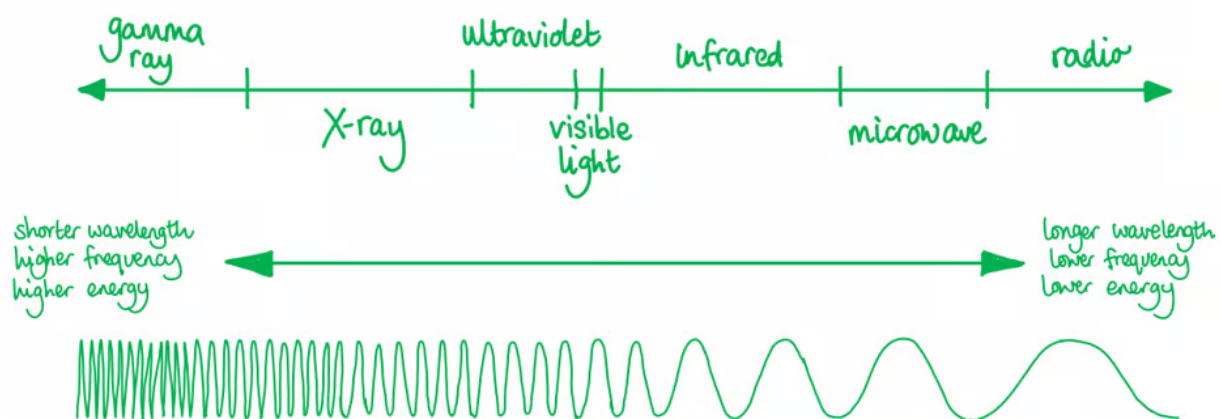
plane only

- Malus's law states that the transmitted intensity of the plane polarised light passing through a rotatable polariser is directly proportional to the square of the cosine of the angle through which the polariser is rotated from the position which gives the maximum intensity.
- Malus's law allows us to calculate the intensity of a beam of plane polarised light, it is given by
 - $I = I_0 \cos^2 \theta$

8

The correct answer is **C** because:

- The electromagnetic spectrum is a term used to describe waves created as a result of vibrations between an electric field and a magnetic field
- The spectrum is made up of :



- This makes Q infrared, and P x-rays

Exam Tip: Make sure to check the labelling for which end is high frequency and low frequency.

9

The correct answer is **D** because:

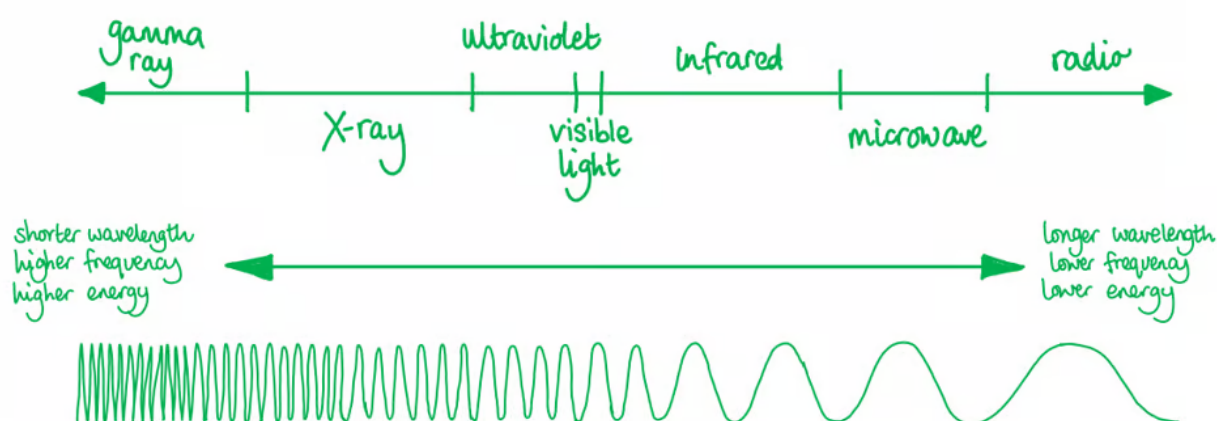
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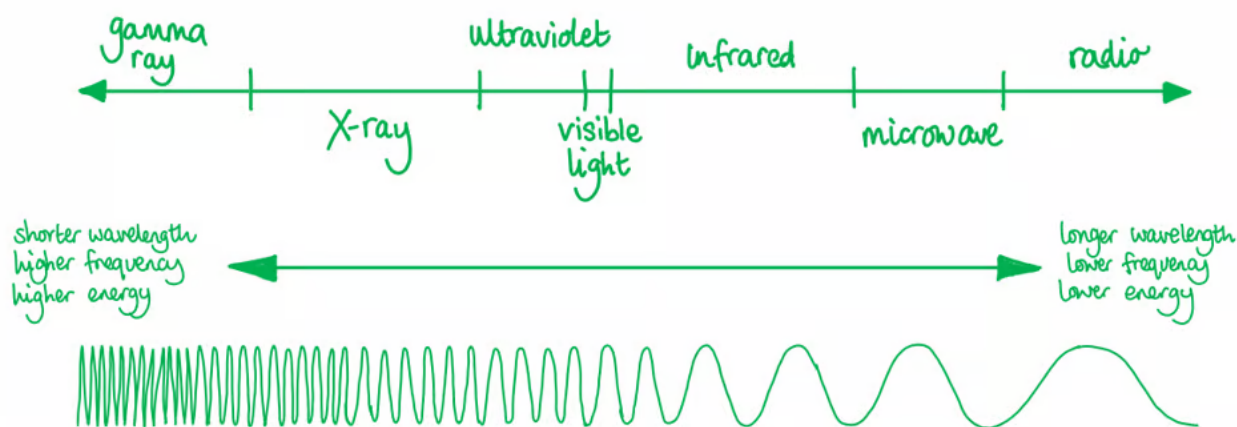
- This makes Q infrared, and P x-rays

Exam Tip: Make sure to check the labelling for which end is high frequency and low frequency.

9

The correct answer is **D** because:

- The electromagnetic spectrum is a term used to describe waves created as a result of vibrations between an electric field and a magnetic field.
- The electromagnetic spectrum is made up of :



- The speed of all EM waves in a vacuum is $3.0 \times 10^8 \text{ m s}^{-1}$
- Electrons do not form part of the EM spectrum, and the speed that they travel in a vacuum is roughly one one-hundredth the speed of light

All the other options are part of the electromagnetic spectrum so all travel at the same speed in a vacuum

10

The correct answer is **B** because:

- The range of wavelength of visible light is typically 400 – 700 nm
- If we take the average value of this range: $550 \text{ nm} = 550 \times 10^{-9} \text{ m}$
- The number of wavelengths in one metre is equal to
 - $\frac{1}{550 \times 10^{-9}} = 1.8 \times 10^6$
- Therefore, there are around 10^6 wavelengths of visible light in 1 m

A is incorrect because this corresponds to a wavelength of $\frac{1}{10^4} = 10^{-4}$ which corresponds to **infra-red** light

C is incorrect because this corresponds to a wavelength of $\frac{1}{10^8} = 10^{-8}$ which corresponds to **ultraviolet** light

D is incorrect because this corresponds to a wavelength of $\frac{1}{10^{10}} = 10^{-10}$ which corresponds to **x-rays**